

Inverse Reinforcement Learning

Question 1. When using Maximum Entropy IRL with high-dimensional function approximation (e.g., neural networks), one key bottleneck is:

- a. The inability to compute gradients with respect to ψ
- b. **The need to differentiate through a complex partition function or to run repeated forward-backward algorithms**
- c. The requirement that all expert trajectories be perfect
- d. The immediate convergence toward a uniform distribution over all trajectories

Question 2. Why is Maximum Entropy IRL used?

- a. To ensure the reward function is constant across all states
- b. **Because the method aims to retain as much randomness as possible, subject to matching expert statistics**
- c. Because it cannot handle suboptimal demonstrations
- d. Because a Dirac delta distribution over the best path is always infeasible

Question 3. What is a common challenge when performing IRL?

- a. There is always a single unique reward function consistent with the expert's behavior
- b. The learned reward is trivial to evaluate without further policy optimization
- c. **Expert demonstrations may be suboptimal, complicating reward inference**
- d. IRL does not require knowledge or assumptions about the environment's dynamics

Question 4. Compared to Imitation Learning (IL), a unique aspect of IRL is that IRL:

- a. **Aims to learn the expert's reward rather than just copying actions**
- b. Only works in deterministic environments
- c. Discards the idea of a reward function entirely
- d. Ignores expert demonstrations

Question 5. In Maximum Entropy IRL, the probability of a trajectory τ is written as:

$$p_\psi(\tau \mid \mathcal{O}_{0:T}) \propto p(\tau) \exp(r_\psi(\tau))$$

What is the role of $p(\tau)$ in this expression?

- a. It enforces that all trajectories with the same reward get zero probability
- b. **It encodes the environment’s dynamics or “how likely” the raw trajectory is, independent of reward**
- c. It ensures that the partition function is always 1
- d. It guarantees that expert demonstrations appear with higher probability than random trajectories by default

Question 6. Which statement correctly describes a slack variable ζ in maximum margin IRL?

- a. **It controls how strictly the constraints must be satisfied, allowing the expert to be suboptimal**
- b. It ensures the maximum margin has infinite size
- c. It forces the expert’s trajectories to always have higher reward
- d. It is irrelevant in IRL and only used in supervised SVMs

Question 7. In IRL, we typically start with:

- a. **A set of expert demonstrations (trajectories)**
- b. A fully known reward function from the environment
- c. A labeled dataset with correct actions for each state
- d. No knowledge about the environment’s dynamics

Question 8. Consider the gradient of the Maximum Entropy IRL objective:

$$E_{\tau \sim \pi^*} [\nabla_\psi r_\psi(\tau)] - E_{\tau \sim p^\psi(\tau \mid \mathcal{O}_{0:T})} [\nabla_\psi r_\psi(\tau)]$$

Conceptually, what does each term represent?

- a. The first term decreases the expert’s reward, the second increases current policy’s reward
- b. Both terms push the reward parameters in the same direction
- c. **The first term reinforces expert trajectories; the second penalizes the policy’s own trajectories**
- d. They both approximate the state visitation frequencies

Question 9. When optimizing the log-likelihood in Maximum Entropy IRL, what makes this problem computationally challenging?

- a. **The partition function Z involves a sum over all possible trajectories, which can be extremely large**
- b. The gradient of the likelihood does not depend on ψ
- c. Each demonstration is guaranteed to be optimal
- d. The objective always converges in one iteration

Question 10. In feature matching IRL with a linear reward $r_\psi(s, a) = \psi^\top f(s, a)$, which statement is true?

- a. The reward is assumed to be a non-linear function
- b. **We aim to match the expert's feature expectations $\mu(\pi)$**
- c. We never need to solve an MDP to update ψ
- d. The reward depends only on the next state transition

Question 11. The structured max-margin IRL formulation uses a loss term $D(\pi_{\text{expert}}, \pi)$ in:

$$\psi^\top \mu(\pi_{\text{expert}}) > \max_{\pi} (\psi^\top \mu(\pi) + D(\pi_{\text{expert}}, \pi))$$

What is the purpose of $D(\pi_{\text{expert}}, \pi)$?

- a. It ensures that only the final reward function matters
- b. **It differentiates between policies that are “almost” like the expert versus those that are very different**
- c. It nullifies any margin gained by suboptimal policies
- d. It speeds up the solver by removing constraints

Question 12. In the maximum margin approach to IRL (inspired by SVMs), what is the primary objective?

- a. To find the widest margin in the raw observations of states
- b. **To maximize the margin between the expert's feature expectations and those of any other policy**
- c. To minimize the number of slack variables to zero
- d. To guarantee that the expert policy is identical to all alternative policies

Question 13. Which of the following is a primary goal of Inverse Reinforcement Learning?

- a. **Recover the reward function from expert demonstrations**
- b. Perform clustering on reward signals
- c. Directly mimic the expert's actions without any reward model
- d. Randomize the learned policy to increase exploration